Simple Linear Regression: Problem Set 1

November 6, 2025

Instructions

This document contains derivation-based questions on Simple Linear Regression (SLR) followed by complete solutions. Each question has a theoretical part and a small numerical example (same dataset used throughout):

Observation	X	Y
1	1	2
2	2	4
3	5	5
4	6	4
5	6	7

Questions

Question 1: Least Squares Estimates

(a) Derive the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ by minimizing

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2.$$

(b) Show that $\sum_{i=1}^{n} \hat{\varepsilon}_i = 0$, where $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$.

(c) Compute $\hat{\beta}_0$ and $\hat{\beta}_1$ for the dataset above.

Question 2: Expectation and Variances

(a) Show $E[\hat{\beta}_1] = \beta_1$ and $E[\hat{\beta}_0] = \beta_0$ under the standard SLR assumptions.

(b) Prove $\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}, \qquad \mathrm{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \right).$

(c) Using the dataset and assuming $\sigma^2 = 0.25$, compute numerical values for the variances and standard errors.

Question 3: Covariance between Estimators

(a) Show that $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{X},\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$.

(b) Give an intuitive explanation for the sign of this covariance when $\bar{X} > 0$.

(c) Compute the numerical value using the dataset and $\sigma^2 = 0.25$.

Question 4: Estimator of Error Variance

(a) Show that the unbiased estimator of σ^2 is $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$.

(b) Prove that $E[\hat{\sigma}^2] = \sigma^2$.

(c) For the dataset, compute $\hat{Y}_i$, residuals $\hat{\varepsilon}_i$, and $\hat{\sigma}^2$.

Question 5: ANOVA Decomposition and R^2

(a) Show algebraically that $\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$.

(b) For the dataset compute Total SS (above first component), Regression SS (second component) and Error SS (third component), and verify the identity.

Question 6: Testing Normality of Residuals

(a) Explain why the assumption of normally distributed errors is important for inference but not for obtaining the least squares estimators.

(b) Describe two graphical methods for assessing the normality of residuals.

(d) If the residuals appear non-normal, discuss two possible remedial measures.

Question 7: Constant Variance Assumption (Homoscedasticity)

(a) In the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad E[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2,$$

The variance of ϵ remains same for all i, this assumption is called *homoscedasticity.*

(b) Explain in words why the assumption of equal variance of errors is important for valid statistical inference on β_0 and β_1.

(c) What is the effect of *heteroscedasticity* (non-constant variance) on the OLS estimators? Do they remain unbiased? Are they still efficient?

(d) Describe one graphical method for checking the constant variance assumption using regression residuals.

Solution to question 7:

(a) Homoscedasticity means that the variance of the error term is the same for all values of the predictor:

$$\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2, \quad \forall i.$$

In words, the spread of the residuals does not depend on X.

(b) If the variance of the errors changes with X, the estimated standard errors of $\hat{\beta}_0$ and $\hat{\beta}_1$ will be incorrect, leading to invalid t-tests and confidence intervals (to be studied in statistical inference, next semester).

(c) The OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ remain *unbiased* under heteroscedasticity, but they are no longer the *best linear unbiased estimators* (BLUE). The Gauss–Markov theorem no longer holds, so the estimators lose efficiency (to be studied later in detail).

(d) A common graphical check is the *residuals vs. fitted values plot.* If the spread of residuals is roughly constant across all fitted values, homoscedasticity is plausible. If the residuals fan out or narrow as fitted values increase, it indicates heteroscedasticity.